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Expt.No - 11(A)

Title - a)Addition of Two Images ¶

b)Subtraction of Two Images

```
In [5]: 1 import cv2
2 import matplotlib.pyplot as plt
3
4 image1 = cv2.imread('resizedImage1.jpg')
5 image2 = cv2.imread('resizedImage2.jpg')
6
7 if image1.shape != image2.shape:
8     print("Error: Images must have the same dimensions!")
9     exit()
10
11 added_image = cv2.add(image1, image2)
12
13 subtracted_image = cv2.subtract(image1, image2)
14
15 plt.figure(figsize=(10, 10))
16
17 plt.subplot(2, 2, 1)
18 plt.imshow(cv2.cvtColor(image1, cv2.COLOR_BGR2RGB))
19 plt.title("Image 1")
20 plt.axis('off')
21
22 plt.subplot(2, 2, 2)
23 plt.imshow(cv2.cvtColor(image2, cv2.COLOR_BGR2RGB))
24 plt.title("Image 2")
25 plt.axis('off')
26
27 plt.subplot(2, 2, 3)
28 plt.imshow(cv2.cvtColor(added_image, cv2.COLOR_BGR2RGB))
29 plt.title("Added Image")
30 plt.axis('off')
31
32 plt.subplot(2, 2, 4)
33 plt.imshow(cv2.cvtColor(subtracted_image, cv2.COLOR_BGR2RGB))
34 plt.title("Subtracted Image")
35 plt.axis('off')
```

```
Out[5]: (-0.5, 199.5, 199.5, -0.5)
```

Image 1



Image 2



Added Image



Subtracted Image

